Strong Readouts, Local Levers: A Steering Audit of Gemma-3-4B-IT

Anonymous Authors[1]

Abstract

Predictive internal signals are often treated as natural targets for steering language models, but their reliability as intervention handles remains unclear. We test this assumption in Gemma-3-4B-IT. In a cross-family FaithEval comparison, sparse autoencoder (SAE) features and magnitude-ranked H-neurons achieve similar held-out detection quality (AUROC 0.848 vs. 0.843), yet only H-neurons produce a reliable compliance dose-response (+2.1 pp/α [1.4, 2.8]; neuron-minus-SAE gap excludes zero). When steering works, it remains surface-local: ITI improves TruthfulQA answer selection but reduces TriviaQA bridge accuracy, most often through wrong-entity substitution. Measurement choices also change the conclusion: truncation and scoring granularity alter the apparent jailbreak result, and a held-out evaluator audit reveals rubric-dependent disagreement. We organize these findings as a four-stage audit framework (measurement, localization, control, and externality) and argue that strong readouts are insufficient evidence for good steering targets.

1. Introduction

A predictive internal signal (a neuron, feature, or direction that reliably discriminates between behavioral categories on held-out data) is a tempting steering target. If a feature predicts whether a model will hallucinate or comply harmfully, it is natural to expect that amplifying or suppressing it will steer behavior. This heuristic underlies much recent work, from representation engineering and activation addition to inference-time intervention and neuron-level steering (Zou et al., 2023; Turner et al., 2023; Li et al., 2023; Gao et al., 2025; Arditi et al., 2024).

The heuristic sometimes works. Refusal-mediating directions produce reliable refusal modulation (Arditi et al., 2024), and hallucination-associated neurons can shift over-compliance (Gao et al., 2025). But the heuristic also sometimes fails, and the failure modes have received less systematic attention than the successes.

This paper tests the readout-to-steering heuristic empirically in Gemma-3-4B-IT (Google DeepMind, 2025). We find repeated dissociations between four stages that are routinely conflated:

1. **Measurement.** Can we trust the evaluation? Truncation and scoring granularity changed the scientific conclusion, while the evaluator audit exposed residual rubric dependence (§5).
2. **Localization.** Does the readout identify causally relevant components? SAE features matched H-neurons on detection quality but did not translate into useful control (§3).
3. **Control.** Does intervention produce the intended behavioral change? Effects were narrow and task-local (§4).
4. **Externality.** Does the effect transfer without causing harm? Wrong-entity substitution was the most common manually coded failure mode on a held-out bridge benchmark (§4).

Contributions. (1) A FaithEval comparison showing that similar held-out readout quality yields sharply different steering outcomes. (2) An externality analysis showing that answer-selection gains do not transfer to generation, with wrong-entity substitution as the most common manually coded failure mode. (3) A four-stage audit framework for evaluating mechanistic intervention claims without collapsing measurement, localization, control, and externality into one inference.

Study orientation. This paper is a single-model comparative case study in Gemma-3-4B-IT (Google DeepMind, 2025). We compare multiple intervention families (neuron scaling, SAE feature steering, and inference-time intervention via attention heads) across contextual faithfulness

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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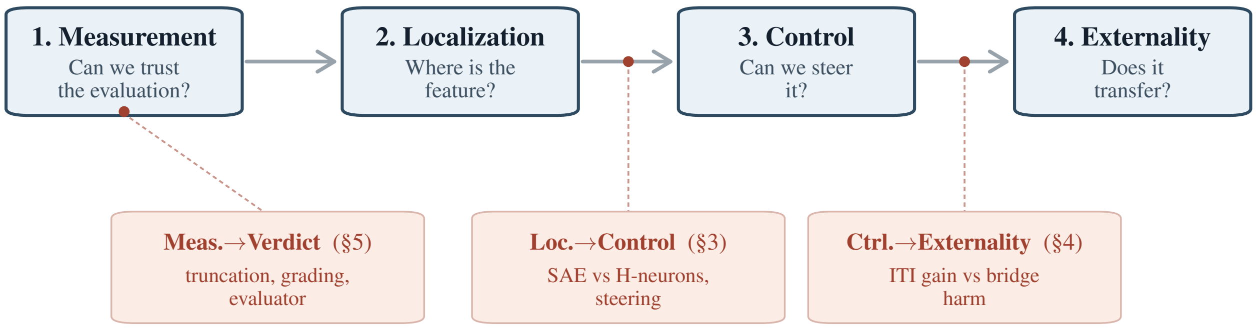

Figure 1. The Four-Stage Interpretability Scaffold and three empirical gates. Stages run left to right; each transition is a question the next stage assumes has been answered. Three gates mark where the readout-to-steering heuristic broke in our case study: one *within-stage* (Measurement→Verdict, §5: truncation and grading shifted the verdict, while the evaluator audit exposed residual rubric dependence) and two *inter-stage* (Localization→Control, §3; Control→Externality, §4). Within-stage gates tether to a stage box; inter-stage gates tether to the transition arrow.

(FaithEval (Ming et al., 2025)), answer selection (TruthfulQA (Lin et al., 2022)), open-ended factual generation (TriviaQA (Joshi et al., 2017), SimpleQA Verified (Haas et al., 2025)), domain QA (BioASQ (Tsatsaronis et al., 2015)), and jailbreak settings (JailbreakBench (Chao et al., 2024)). All claims are surface-specific: FaithEval is a context-faithfulness / anti-compliance surface, TruthfulQA MC constrained answer selection, TriviaQA bridge and SimpleQA Verified open-ended factual generation, and JailbreakBench a harmfulness-evaluation surface. Within that scope, the paper asks three bounded empirical questions: (1) Is held-out readout quality enough to select steering targets? (§3) (2) Do answer-selection gains transfer to open-ended factual generation? (§4) (3) Do plausible measurement choices leave the jailbreak verdict stable? (§5) These are surface-specific questions, not claims of a new steering method or a general failure of detector-based targets.

2. Related Work

Foundational probe critiques establish that high readout accuracy can arise for reasons the model does not functionally rely on (Hewitt & Liang, 2019; Elazar et al., 2021; Kumar et al., 2022). The linear representation hypothesis (Park et al., 2024) makes the complementary positive prediction that concepts encoded as linear directions should support both detection and control, which the present paper probes empirically. Hase et al. (2023) provide a close editing precedent: better localization need not predict better control.

The steering lineage motivating this audit runs from representation engineering and activation addition (Zou et al., 2023; Turner et al., 2023) through ITI (Li et al., 2023). Within SAE-based interventions, foundational feature work (Cunningham et al., 2023) and the Gemma Scope release (Lieberum et al., 2024) make sparse features available at scale, while target-selection studies show that not all salient features steer well: high-scoring SAE features need not control behavior (Arad et al., 2025), detection and steering need not be jointly optimized (Wu et al., 2025), Wang et al. (2026) report only a weak positive association between SAE interpretability and steering utility across 90 SAEs (Kendall's $\tau_b \approx 0.298$), and Bhalla et al. (2024) describe a predict/control discrepancy for interpretability-based interventions.

Transfer and reliability work narrows the claim space further. Li et al. (2023) already showed that ITI's multiple-choice gains can diverge from generative gains; Tan et al. (2024) study the brittleness and input sensitivity of steering vectors; Pres et al. (2024) show that behavior-steering evaluations can be format-dependent; and Opiełka et al. (2026) show that causally effective vectors can still be surface-specific. Positive counterexamples remain important: refusal directions and targeted multi-attribute steering can work when the behavioral target is better matched to the intervention (Arditi et al., 2024; Nguyen et al., 2025).

Measurement fragility is a parallel reviewer concern. JailbreakBench (Chao et al., 2024) provides the benchmark surface used here; StrongREJECT (Souly et al., 2024) showed that binary scoring can overestimate jailbreak effectiveness; Chen & Goldfarb-Tarrant (2025) and Eiras et al. (2025) show that evaluator artifacts and rubric choices matter; and Huang et al. (2025) argue that guideline-conditioned judging can materially change jailbreak conclusions.

Concurrent SAE-centric analyses (Arad et al., 2025; Wang et al., 2026) and synthetic-benchmark evaluations (Wu et al., 2025) make detection/control divergence increasingly clear within single-family settings. The core contribution here is a cross-representational FaithEval comparison: neurons and SAE features are evaluated on the same real behavioral surface with closely matched held-out readout quality. Two supporting additions broaden that case study. First, on TriviaQA bridge generation we provide a frozen-protocol behavioral diagnosis of the transfer failure, where wrong-entity substitution is the most common manually coded right-to-wrong pattern rather than simple suppression. Second, we show that truncation and scoring granularity can change the verdict for the same representation-engineering intervention outputs, while a held-out evaluator audit reveals residual rubric dependence. We synthesize these observations in a four-stage audit framework (measurement, localization, control, externality).

3. Similar Readout Quality Does Not Guarantee Control

3.1. The Readouts Are Real

The intervention targets were selected through held-out predictive readouts. A sparse H-neuron probe following Gao et al. (2025) identified 38 positive-weight neurons out of 348,160 feed-forward neurons and achieved AUROC 0.843 (accuracy 76.5%, $n_{\text{test}} = 780$). A parallel L1 probe on Gemma Scope 2 SAE activations (Lieberum et al., 2024) selected 266 positive-weight features across 10 layers and achieved AUROC 0.848 (accuracy 77.2%, $n_{\text{test}} = 782$). These are real held-out signals, though individual-weight interpretation remains fragile (Appendix A).

3.2. Comparable Readouts, Divergent Control

Both methods achieve similar held-out detection quality, and we evaluate both interventions on the same 1,000 FaithEval items (Ming et al., 2025). The 38 probe-selected neurons were scaled multiplicatively across $\alpha \in \{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, with $\alpha = 1.0$ as the no-op baseline. The 266 SAE features were scaled through an encode-modify-decode intervention on the same grid.

The neuron intervention showed a clear dose-response: compliance slope $+2.09$ pp/α $[1.38, 2.83]$, with perfect Spearman monotonicity ($\rho = 1.0$). Relative to the no-op baseline, compliance at $\alpha = 3.0$ increased by $+4.5$ pp $[2.9, 6.1]$. Across five unconstrained and three layer-matched random-neuron sets, all eight random seeds produced null slopes (maximum $+0.21$ pp/α); every paired neuron-minus-random difference excluded zero.

The SAE intervention did not produce the same behavior. The full-replacement H-feature slope was $+0.16$ pp/α $[-0.51, 0.84]$, with no monotonic trend ($\rho = 0.18$). A delta-only architecture that cancels reconstruction error confirmed the null: delta-only H-feature slope $+0.12$ pp/α, delta-only random-feature slope -0.09 pp/α, both indistinguishable from zero. The neuron baseline on the same three-alpha subset remained $+2.12$ pp/α.

The paired neuron-minus-SAE slope difference was $+1.93$ pp/α $[+0.94, +2.92]$ (permutation $p < 0.001$). On the same 1,000 FaithEval items, similar held-out readout quality did not predict intervention utility: H-neurons and SAE features showed comparable detector performance, but only the neuron intervention produced a robust behavioral dose-response.

Within-SAE target-selection ablation. To test whether the SAE null reflects a poor target-selection rule rather than the SAE family, we ran an on-pool ablation on the same 509-feature candidate set using a single $\alpha = 0$ delta-only operator (feature zeroing) on a frozen $160/840$ validation/test FaithEval split. The first ablation compared top-266 readout features, top-266 prompt-end validation-utility features, and ten layer-matched activation-weighted random zero-weight controls; all produced null compliance deltas on the $n = 840$ test split (all paired 95% CIs include zero; point estimates ≤ 0.72 pp). On the anti-compliance log-prob margin, readout-selected features shifted the margin in the *wrong* direction (+0.92 nats $[+0.61, +1.23]$ vs. no-op), while prompt-end utility-selected features reduced it by -0.76 nats $[-1.08, -0.42]$; the paired utility-minus-readout delta was -1.68 nats $[-2.17, -1.19]$. Because both sides of that contrast share the same $\alpha = 0$ delta-only operator, any path artifact cancels; a dedicated path-drift control ($\alpha = 0$ on a single provably-dead zero-weight feature) independently moved the margin by only $+8.2 \times 10^{-8}$ nats $[-2.4 \times 10^{-3}, +2.1 \times 10^{-3}]$. A later answer-span selector on the same pool added a narrower finding: it reduced its native first-3-token answer-text margin (-0.33 nats $[-0.61, -0.06]$ vs. no-op), but left compliance null and underperformed the prompt-end utility selector on the original prompt-end margin ($+0.46$ nats $[+0.16, +0.75]$). Within the same candidate pool, readout-, prompt-end-utility-, and answer-span-utility criteria therefore induce metric-specific margin effects, and no rule reaches the accuracy endpoint (Appendix D).

3.3. Synthesis

H-neurons *did* steer FaithEval compliance, and specificity was confirmed against eight matched random controls. The refusal-direction literature (Arditi et al., 2024) and targeted multi-attribute steering (Nguyen et al., 2025) are positive cases. The narrower claim is that readout quality alone is an unreliable heuristic for predicting *when* detection-based targets will steer behavior. SAE features matched H-neurons on the quantity most commonly used to justify steering (held-out AUROC) and failed on the downstream behavioral test.

A natural objection is that this comparable-readout design confounds representational basis with readout quality: SAE features and neurons differ in operator form, layer coverage, and feature granularity. The within-SAE ablation above partly closes that gap: with operator form and the available extraction layers held fixed, readout-, prompt-end-utility-, and answer-span-utility selectors produce metric-specific margin effects, and none reaches the accuracy endpoint. The remaining basis confound narrows the across-family inference, but readout quality alone did not suffice to predict steering utility in either the cross-representational comparison or within the SAE pool. We discuss this limitation further in §6.

4. Control Is Surface-Local and Can Externalize

4.1. Positive Results Are Task-Local

Before testing transfer, we establish that detector-selected interventions can steer behavior on their primary surfaces. H-neuron scaling (38 magnitude-ranked neurons) produces clear, dose-dependent effects on compliance-adjacent surfaces. On FaithEval, compliance increases by $+4.5$ pp $[2.9, 6.1]$ above the no-op baseline at $\alpha = 3.0$ (slope $+2.09$ pp/α), confirmed by 8-seed negative controls showing flat random-neuron baselines (mean slope $+0.02$ pp/α). On FalseQA (Hu et al., 2023) ($n = 687$), the same neurons show a dose-response slope of $+1.62$ pp/α $[0.52, 2.74]$.

However, the same 38 neurons show no robust net accuracy effect on BioASQ factoid QA (Tsatsaronis et al., 2015): -0.06 pp $[-1.5, 1.4]$ on a well-powered sample ($n = 1{,}600$, MDE $\sim$2 pp), despite substantially perturbing answer style in 1,339 of 1,600 responses. The flat accuracy endpoint coexists with substantial output perturbation, suggesting that behavioral activity and alias-accuracy movement come apart on this surface. The intervention modulates compliance-adjacent behavior but does not improve domain-specific factual accuracy under the alias metric.

4.2. ITI: Answer Selection vs. Open-Ended Generation

ITI with truthfulness directions fit from TruthfulQA (Lin et al., 2022) following Li et al. (2023) improves answer selection. At $\alpha = 8.0$, it yields $+6.3$ pp MC1 $[3.7, 8.9]$ and $+7.49$ pp MC2 $[5.28, 9.82]$ on held-out TruthfulQA folds, with 61 incorrect→correct flips against 20 correct→incorrect flips.

This MC/generation divergence replicates the finding of Li et al. (2023) in LLaMA/Alpaca; here, SimpleQA Verified serves as a supporting stress test, while the locked bridge benchmark below is the main externality result. The gain does not transfer to open-ended factual generation. On a local copy of the 1,000-question SimpleQA Verified benchmark (Haas et al., 2025), graded with the benchmark's verified autorater, and with a prompt that removes the explicit escape hatch, ITI at $\alpha = 8.0$ reduces correct-answer rate from 4.6% $[3.5, 6.1]$ to 2.8% $[1.9, 4.0]$ (paired $\Delta = -1.8$ pp $[-3.1, -0.6]$). The attempt rate collapses from 99.7% to 67.0%, while precision among attempted answers remains stable ($\sim$4%), suggesting epistemic caution rather than factual improvement. Answer-selection success is therefore not evidence for generation-level truthfulness improvement, consistent with Pres et al. (2024) and Opiełka et al. (2026).

4.3. Bridge: Wrong-Entity Substitution

The TriviaQA bridge benchmark (Joshi et al., 2017) (held-out test set, $n = 500$, baseline adjudicated accuracy 45.0%) provides the most informative externality result because it reveals a specific behavioral failure mode, not just a score drop. The test set was evaluated under a one-shot frozen protocol: all parameters were locked from Phase 2 dev results, and no parameters were tuned on test data.

The intervention is active, not simply suppressive. At $\alpha = 8.0$, the baseline TruthfulQA-sourced ITI variant reduces adjudicated accuracy by -5.8 pp $[-8.8, -3.0]$ (CI excludes zero), with 43 right-to-wrong flips against 14 wrong-to-right rescues (net -29, McNemar $p = 0.0002$). Both the primary (adjudicated) and floor (deterministic) accuracy deltas exclude zero, ruling out a grading artifact.

The observed harm is not primarily refusal: NOT_ATTEMPTED counts rise from 2 to 8, a small 1.2 pp effect. Formal refusal accounts for 0 of 43 R→W flips under the dual-rated coding. The dominant damage is factual corruption, not silence.

Wrong-entity substitution is the most common dual-rated failure mode. Entity-level factuality work treats wrong-entity substitution as a recognizable neighboring error family in generation (Nan et al., 2021); we use that literature only as adjacent taxonomy, not as prior evidence for this exact steering failure mode. Each of the 57 discordant test cases (43 R→W + 14 W→R) was coded by the first author and a blinded LLM judge (`gpt-4o-2024-11-20`, temperature 0, strict JSON schema) against a pre-registered four-category rubric whose decision rules were committed before any test label was appended. Raw agreement was 96.5% (Cohen's $\kappa = 0.90$, Gwet's AC1 = 0.96); the two disagreements were resolved under the pre-committed rule. Of the 43 R→W flips, 31 fell in the wrong-entity substitution category: the model replaces a correct factual answer with a different entity from the same semantic neighborhood ($31/43 = 72\%$; 95% CI $[57, 83]$; Table 1). The remaining 12 flips split into evasion/factual denial (9/43, 21%), answer dilution (3/43, 7%), and formal refusal (0/43). This profile is consistent with a coarse behavior-level reweighting

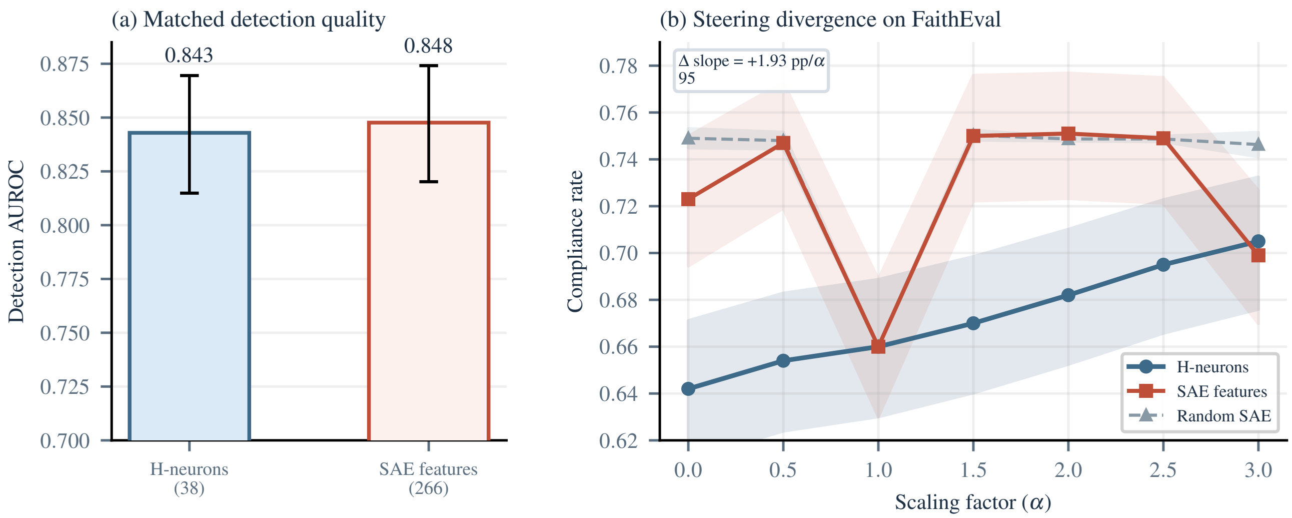

Figure 2. The anchor result: FaithEval comparable readouts, divergent control. (a) H-neurons and SAE features have similar held-out detection AUROC. (b) Only H-neurons produce a dose-dependent compliance increase; SAE features and random SAE features are both flat. The neuron-minus-SAE slope difference ($+1.93$ pp/α, 95% CI excludes zero) confirms the dissociation.

toward nearby but wrong candidates rather than suppression of generation; we treat it as a behavioral diagnosis, not a claim about the exact internal circuit mechanism. A teacher-forced gold-vs-wrong log-likelihood-margin check (Appendix F) has the same directional R→W/W→R sign, but no substitution-specific signature: non-substitution flips show larger compression than substitution flips.

The 14 wrong-to-right rescues show that the intervention is behaviorally active in both directions; all 14 are themselves coded as wrong-entity substitutions (Wilson 95% CI $[78.5, 100]$), reinforcing the reweighting reading, and on this surface the damage rate is larger than the rescue rate.

5. Measurement Choices Changed the Conclusion

The preceding sections established that detection quality does not predict steerability (§3) and that successful steering is narrow in scope (§4). Both conclusions rest on behavioral measurements (jailbreak harmfulness rates, severity scores, and generation-surface accuracy) that are themselves products of evaluation choices. In this section we show that those choices are part of the scientific result: truncation and scoring granularity shifted what we would have concluded about whether a given intervention worked on the same outputs, while the evaluator audit exposes residual rubric dependence on a held-out set. The general point that evaluator choices matter is established (§2); the specific finding here is that same-output measurement choices can reverse the conclusion about a representation-engineering intervention in the RepE sense (Zou et al., 2023), even when two held-out evaluators tie in aggregate accuracy.

We organize the case study around the H-neuron jailbreak scaling experiment (38 probe-selected neurons, $\alpha \in \{0, 1, 1.5, 3\}$, $n = 500$ per condition), because its moderate effect size makes it sensitive to every measurement decision we examine.

Truncation hides downstream content. Short generation caps (e.g., 256 tokens) can systematically bias intervention evaluations when steering changes response structure. Gemma-3-4B-IT often begins jailbreak responses with a refusal preamble and only later emits substantive harmful content. Short caps therefore preferentially capture the refusal-looking prefix and hide the downstream payload, overestimating intervention effectiveness. Full-generation scoring is therefore required for the kinds of jailbreak claims made in this paper.

Scoring granularity changes the verdict. The same H-neuron jailbreak outputs look different under binary and graded evaluation. A GPT-4o binary harmful/safe judge shows a $+3.0$ pp endpoint shift ($\alpha = 0 \rightarrow 3$, 152/500 $\rightarrow$ 167/500), with a confidence interval that includes zero. Under binary evaluation alone, the intervention looks weak and non-decisive.

A graded harmfulness rubric (CSV-v2) applied to the same outputs recovers a different result. The H-neuron strict harmfulness slope is $+2.30$ pp/α $[+0.99, +3.58]$, while a matched random-neuron control is flat at -0.47 pp/α $[-1.42, +0.47]$. The slope difference is $+2.77$ pp/α $[+1.17, +4.42]$ with permutation $p = 0.013$. Binary scoring washes out this signal because the effect largely lives in the movement of borderline cases into clearer refusal or clearer compliance.

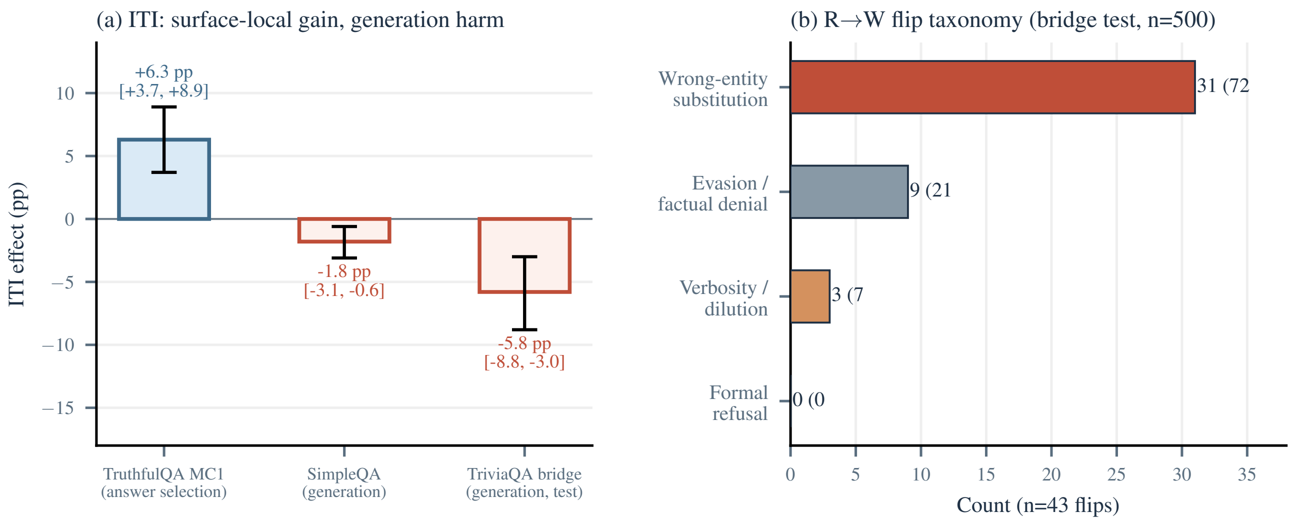

Figure 3. Surface-local control and bridge failure modes. (a) ITI improves TruthfulQA MC1 but harms both generation surfaces (CIs exclude zero). (b) Among the 43 right-to-wrong flips on the TriviaQA bridge test set, wrong-entity substitution is the most common dual-rated failure mode (72%; 95% CI $[57, 83]$), not refusal.

This is the section's core granularity result: binary evaluation suggests a weak or null effect; graded evaluation shows a positive dose-response.

Evaluator dependence is part of the result. On a contamination-free 50-record holdout, four GPT-4o-based evaluators yield the accuracies in Table 2. No pairwise difference reaches statistical significance at $n = 50$. CSV-v3 and StrongREJECT-GPT-4o both reach 96.0% holdout accuracy with identical error sets. We retain CSV-v3 for its richer outcome taxonomy, not for accuracy superiority.

The residual disagreement is informative because it clusters on a recognizable response type: refuse-then-comply outputs. StrongREJECT's formula zeroes the final score when `refused = 1`, even if later text contains specific harmful substance. We therefore treat evaluator disagreement as construct pluralism rather than noise that can be solved by picking one preferred judge (Eiras et al., 2025).

6. A Four-Stage Audit Framework

We distill the case study evidence into a practical audit for evaluating mechanistic intervention claims. The audit has four stages: measurement, localization, control, and externality. They are not a taxonomy. They are an empirical observation about where intervention claims break in practice:

- **Measurement→Verdict.** Before deciding whether an intervention worked, one must trust the evaluation that defines the behavioral surface. In our jailbreak setting, truncation and binary-versus-graded scoring altered the conclusion about the same outputs, while the evaluator holdout exposed residual construct mismatch (§5).
- **Localization→Control.** A feature that predicts behavior on held-out data need not causally control that behavior when perturbed. SAE features matched H-neurons on detection quality, yet did not translate into useful control (§3).
- **Control→Externality.** An intervention that succeeds on one surface may fail or cause harm on a nearby surface. ITI improved TruthfulQA MC1 but reduced open-ended factual accuracy, where wrong-entity substitution was the most common manually coded failure mode (§4).

Each stage transition is a distinct empirical claim. Passing one does not license claims about the next. Table 3 summarizes the minimum evidence required at each gate.

In our reading of the recent steering literature, target-surface control is reported more often than matched negative controls, cross-surface tests, or evaluator audits. In our case study, each omitted gate would have licensed a claim that the full audit rejects: without measurement audits, the jailbreak intervention looks null; without externality checks, the ITI intervention looks beneficial. The checklist is an empirical codification of the points where our own conclusions changed.

6.1. Five Recommendations

R1: Do not treat held-out readout quality as sufficient target-selection evidence. Metrics such as AUROC, accuracy, and probe coefficient magnitude are insufficient on

Table 1. Wrong-entity substitution examples from the TriviaQA bridge test set (E0 $\alpha = 8$). The model swaps correct entities for semantically nearby incorrect ones.

Question	Baseline	ITI $\alpha = 8$
Danny Boyle's 1996 "Choose Life" film?	Trainspotting	Slumdog M. (same director)
Other musician with Buddy Holly and the Big Bopper in the Feb. 1959 crash?	Ritchie Valens	J.P. Richardson (same crash)
Family Guy character who moved to Stoolbend, Virginia?	Cleveland Brown	Peter Griffin (same show)
Dickens novel with Merdle, Sparkler, and Tattycoram?	Little Dorrit	Bleak House (same author)
Comic whose June 1938 issue introduced Superman?	Action Comics	Detective Comics (same publisher)

Table 2. Holdout evaluator accuracy (contamination-free split, $n = 50$). Prompt-clustered bootstrap 95% CIs.

Evaluator	Accuracy	95% CI
CSV-v3 (GPT-4o)	96.0%	[90.0, 100.0]
StrongREJECT (GPT-4o)	96.0%	[90.0, 100.0]
CSV-v2 (GPT-4o)	92.0%	[84.3, 98.0]
Binary judge (GPT-4o)	90.0%	[80.0, 98.0]

their own for identifying useful steering targets. Our comparison in §3 shows that similar held-out AUROC can fail to predict intervention utility.

R2: Validate on the behavioral surface you actually care about. Answer-selection benchmarks and open-ended generation benchmarks measure different constructs, and intervention effects do not transfer reliably between them (§4).

R3: Use matched negative controls. When component search is large, post-hoc selection creates a multiple-comparisons problem. Multi-seed random controls are necessary to establish that an effect is component-specific rather than a generic perturbation artifact.

R4: Treat evaluator disagreement as information. When an intervention changes the surface form of outputs, different evaluators may reach different conclusions. This is not noise; it reflects construct mismatch (§5).

R5: Report externality costs as first-class outcomes. Interventions that help on one metric can harm others. Our bridge results show that the harm can be a specific, interpretable failure mode (§4). Cross-surface effects belong next to the headline result, not in the footnotes.

Table 3. Minimum audit for a credible steering claim. A result that clears only one or two gates is a monitoring or localization result, not a steering result.

Gate	Evidence required
Measurement	Pre-specified metric; full-generation scoring; evaluator agreement check if intervention alters output style
Localization	Held-out readout quality; but also: does steering through these components change behavior?
Control	Dose-response on target surface; matched negative control (random component, random direction); effect size with CI
Externality	At least one non-target surface; capability check; report harms and scope conditions explicitly

6.2. Limitations

All experiments use a single model (Gemma-3-4B-IT); the dissociations may not replicate across architectures or scales, though Gao et al. (2025) report qualitatively similar H-neuron compliance effects across six models. The comparable-readout design keeps held-out AUROC close but cannot rule out that some other property of the feature family (operator form, layer coverage) explains the steering divergence. The within-SAE selector ablations reduce the target-selection concern but do not remove the cross-family operator-form confound. SAE layer coverage is partial; a different SAE width or layer selection could yield non-null results. The bridge failure-mode second rater is an LLM judge (GPT-4o), not an independent human coder, so we present the agreement reported in §4 as a sensitivity check rather than strong-form human–human IRR. Jailbreak H-neuron specificity is single-seed ($p = 0.013$); additional seeds are needed. Concurrent work (Wang et al., 2026; Arad et al., 2025; Wu et al., 2025) already argues for interpretability/utility divergence from SAE-centric and benchmark-driven perspectives; we do not claim discovery of detector/control dissociation in general. Our distinctness lies in a stricter audit scaffold and a matched cross-representational comparison on a real behavioral surface. The full limitation inventory is in Appendix G.

7. Conclusion

In Gemma-3-4B-IT, strong internal readouts supported monitoring and sometimes supported control. But their usefulness as steering targets depended on the behavioral surface and the measurement procedure used to judge success. The four-stage audit framework (measurement, localization, control, and externality) summarizes that bounded lesson: passing one gate does not license claims about the next.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically evaluation methodology for safety-relevant model interventions. The wrong-entity substitution failure mode documented here has implications for deployment of steering-based safety mitigations: interventions that appear beneficial on constrained evaluation surfaces may corrupt factual generation in ways that are harder to detect than outright refusal.

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A. Detector Interpretation Audits

The top-weight CETT neuron (L20:N4288) is not stable across regularization settings: it is absent at $C \leq 0.3$, appears at $C = 1.0$, and drops to rank 5 at $C = 3.0$. The extreme weight is better explained by L1 concentration among correlated features than by unique causal importance. Additionally, full-response readouts are strongly length-confounded: response length dominates truthfulness signal by $\sim 3.7\times$ under mean aggregation. These are not threats to the intervention claims (FaithEval uses single-letter extraction), but they justify the narrower wording in the main text.

B. Construct Map

C. FaithEval Detailed Rates

D. SAE Target-Selection Ablation on FaithEval

Section 3 briefly reported a within-SAE target-selection ablation; this appendix provides the numerical detail.

Design. Candidate pool: 509 SAE features with non-zero weight in the frozen FaithEval SAE classifier, drawn from the existing extraction layers $\{0, 5, 6, 7, 13, 14, 15, 16, 17, 20\}$ at $d_{\text{SAE}} = 16\,384$. Stratified $160/840$ validation/test split by $(\text{num\_options}, \text{counterfactual\_key})$, seed 42, with validation $\cap$ test $= \emptyset$. Operator: SAE delta-only ablation at $\alpha = 0$ (encode $\rightarrow$ zero the target feature $\rightarrow$ decode $\rightarrow$ add the delta to the residual stream). Three 266-feature selection rules are compared (readout: top probe weight; prompt-end utility: top validation reduction in the forced-choice anti-compliance margin; answer-span utility: top validation reduction in the first-3-token answer-text margin). The utility and answer-span selectors each receive ten layer-matched zero-weight random seeds, sampled without replacement with weights proportional to validation-token activation rate via Efraimidis-Spirakis keys from a 112 004-feature eligible pool. A single-feature *path-drift control* ($\alpha = 0$ on a layer-20 feature with zero classifier weight *and* zero validation-token activation) isolates the residual contribution of routing through the SAE hook when the targeted feature is provably dead.

Results. Table 6 reports compliance and the two margin endpoints on the frozen $n = 840$ test split.

All selector families produce null compliance deltas (all paired 95% CIs include zero; the answer-span selector's largest point estimate is $+0.95$ pp $[-0.83, +2.74]$ vs. no-op), so the FaithEval accuracy null is not rescued by any selection rule we tried. On the prompt-end anti-compliance

Table 4. Benchmark construct map. Each evaluation surface measures a specific behavioral construct.

Benchmark	Construct	Primary Metric	Main Caution
FaithEval	Context-resistance under anti-compliance prompting	Compliance rate	Credulity lever, not standard truthfulness
TruthfulQA MC1/2	Answer selection under constrained candidates	MC1 accuracy	Does not measure open-ended generation
TriviaQA bridge	Short-form factual generation	Adj. accuracy	Dual-rated failure coding ($\kappa = 0.90$)
JailbreakBench	Harmful compliance under adversarial prompting	Graded harm rate	Binary eval. underpowered (MDE ∼6 pp)
BioASQ	Domain-specific factoid QA (biomedical)	Accuracy	Flat despite behavioral perturbation

Table 5. FaithEval compliance by method and scaling factor ($n = 1{,}000$; Wilson 95% CIs).

α	Neurons	SAE H-feat.	SAE rand.
0.0	64.2%	72.3%	74.9%
0.5	65.4%	74.7%	74.8%
1.0	66.0%	66.0%	66.0%
1.5	67.0%	75.0%	75.0%
2.0	68.2%	75.1%	74.9%
2.5	69.5%	74.9%	74.9%
3.0	70.5%	69.9%	74.6%

margin, readout-selected features shift the margin in the *wrong* direction while prompt-end utility-selected features modestly improve it; the paired utility-minus-readout delta is -1.68 nats $[-2.17, -1.19]$. This within-pool contrast is immune to path drift because both sides share the same $\alpha = 0$ delta-only operator; the explicit path-drift control ($+8.2 \times 10^{-8}$ nats $[-2.4 \times 10^{-3}, +2.1 \times 10^{-3}]$) independently bounds invocation-only code-path drift far below the selection-specific effects.

A strictly-positive-utility augment ($k = 154$) guards against size-match dilution and reduces the margin by -0.72 nats $[-1.04, -0.39]$ vs. no-op, with three layer-matched random-positive seeds yielding paired deltas -0.74 $[-1.10, -0.36]$, -0.81 $[-1.16, -0.46]$, and -0.42 $[-0.75, -0.08]$ nats. The ten-seed matched-random ensemble for the main bundle gives a seed-mean utility-minus-matched-random contrast of -0.90 nats (nested paired bootstrap $[-1.22, -0.56]$; naive seed bootstrap $[-1.17, -0.61]$); 9 of 10 seeds produce a negative paired delta. The answer-span selector also improves over its own ten-seed matched-random null on its native margin (seed-mean -0.53 nats, nested $[-0.81, -0.26]$), but this is within-metric generalization from validation to test: the more conservative cross-metric contrast is mixed, with answer-span selection worse than prompt-end utility selection by $+0.46$ nats $[+0.16, +0.75]$ on the original prompt-end margin. The selector was frozen before any test labels were scored, and the random-null construction, the path-drift control, the 10-seed extension, and the answer-span extension were added sequentially in response to the adversarial audit flagged in our provenance notes.

Reading. Within the same candidate pool, readout-, prompt-end-utility-, and answer-span-utility criteria pick feature sets with metric-specific margin effects, and no rule reaches the accuracy endpoint. The two intervention-aware $k = 266$ sets share 167 features (Jaccard 0.46), so the contrast is best read as evidence that the intervention objective is load-bearing within this pool, not as evidence that either objective discovers a behavioral steering handle. The FaithEval SAE null is therefore not a readout-selection artifact within the available SAE extraction layers (L2 in Table 9). Neither contrast elevates the SAE intervention to a successful steering handle; the margin-level separation between selection rules does not transfer to the behavioral endpoint.

E. Benchmark Power Summary

F. Bridge Margin-Shift Analysis

Section 4 reported the bridge wrong-entity-substitution taxonomy as a behavioral diagnosis. This appendix quantifies the corresponding teacher-forced log-likelihood-margin shifts at the same intervention ($\alpha = 8$, $K = 12$, `first_3_tokens` decode scope) and states what the analysis does and does not support.

Design. For each case we sum per-token gold and wrong-entity logprobs over the first three continuation positions under baseline and ITI conditions, and report the per-case shift $\Delta_{\text{margin}} = \Delta_{\text{ITI}} - \Delta_{\text{base}}$, where $\Delta = \log p(\text{gold}) - \log p(\text{wrong})$. Four cohorts: (A) $n = 31$ right-to-wrong (R→W) wrong-entity substitutions; (B) $n = 12$ R→W non-substitution flips (evasion/denial, dilution); (C) $n = 14$ wrong-to-right (W→R) rescues; (D) $n = 200$ length-matched random wrong-entity controls drawn from unrelated bridge questions. The analysis plan (endpoints, cohort

Table 6. SAE target-selection ablation on FaithEval ($n = 840$). Margin contrasts are vs. no-op; lower margins are better because they reduce the misleading-vs-preferred logprob gap. CIs are 10 000-sample paired bootstrap (95%).

Family	**Compliance**	**Prompt-end margin contrast**	**Answer-span margin contrast**
No-op	66.4%	—	—
Readout-selected (k=266)	66.0%	$+0.92$ $[+0.61, +1.23]$	$+0.65$ $[+0.30, +0.99]$
Prompt-end utility-selected (k=266)	66.2%	-0.76 $[-1.08, -0.42]$	-0.10 $[-0.30, +0.10]$
Answer-span utility-selected (k=266)	67.4%	-0.31 $[-0.73, +0.12]$	-0.33 $[-0.61, -0.06]$
Path-drift (k=1 dead)	66.4%	$+8.2\times10^{-8}$ $[-2.4\times10^{-3}, +2.1\times10^{-3}]$	n/a

Table 7. Benchmark power and minimum detectable effect (MDE) summary.

Benchmark	n	**Slope**	**MDE**
FaithEval	1,000	$+2.09$ pp/α	$\sim$3 pp
FalseQA	687	$+1.62$ pp/α	$\sim$4 pp
JailbreakBench	500	$+2.30$ pp/α	$\sim$5 pp
BioASQ	1,600	flat	$\sim$2 pp

partition, one-sided permutation direction) was sealed *before* the full 257-case run executed; results are reported against that locked tree.

Table 8. Teacher-forced gold-vs-wrong log-likelihood-margin shift on the TriviaQA-bridge discordant set (first-3 tokens, ITI $\alpha = 8$ vs. baseline). Bootstrap 95% CIs, 10 000 resamples, seed 42.

Cohort	n	Δ_{margin}	95% CI
A: R→W substitution	31	-10.16	$[-12.81, -7.60]$
B: R→W non-subst.	12	-40.06	$[-50.09, -30.37]$
C: W→R rescue	14	$+4.73$	$[+2.27, +7.25]$
D: random control	200	-5.05	$[-6.55, -3.60]$

What holds. ITI causally compresses the gold-vs-wrong margin on R→W cases and expands it on W→R rescues, in the direction behavioral outcomes demand. Substitution cases are significantly more compressed than length-matched random wrong-entity controls (A − D $= -5.10$ nats $[-8.22, -2.16]$, one-sided permutation $p = 0.0081$).

What does not hold. Non-substitution R→W flips show *larger* margin compression than substitution flips ($A - B = +29.90$ nats $[+19.91, +40.54]$; the substitution-specific prediction is formally reversed), concentrated but not exclusively at the first-token answer frame (position 0: $B = -21.74$ nats $[-29.48, -14.85]$ vs. $A = -3.49$ nats $[-5.14, -1.97]$). The behavioral wrong-entity-substitution taxonomy ($\kappa = 0.90$) is therefore a reliable description of *what* the model emits, but does not correspond to a distinct margin-shift signature at the log-likelihood level.

This refines rather than contradicts the §4 externality claim: the dominant-failure-mode statement is behavioral and interrater-defensible, and the margin analysis confirms ITI shifts probability mass in the predicted direction but changes the margin-level reading from substitution-specific entity suppression to answer-commitment margin compression on R→W flips in general.

G. Limitation Inventory

Table 9. Key limitations and which claims they constrain.

#	**Limitation**
L1	Single model (Gemma-3-4B-IT); all claims.
L2	Matched-readout confound: the target-selection artifact form is addressed within the SAE pool (readout-, prompt-end-utility-, and answer-span-utility selectors give metric-specific margin effects with a null accuracy endpoint; Appendix D); operator-form differences across families remain untested.
L3	SAE layer coverage is partially addressed (selector searched all probe-nonzero features in the existing extraction layers); a different width or layer family could yield non-null results.
L4	Bridge failure-mode second rater is an LLM judge (GPT-4o), not an independent human coder; raw agreement 96.5% ($\kappa = 0.90$, AC1 = 0.96).
L5	Jailbreak H-neuron specificity is single-seed ($p = 0.013$); seeds 1–2 pending.